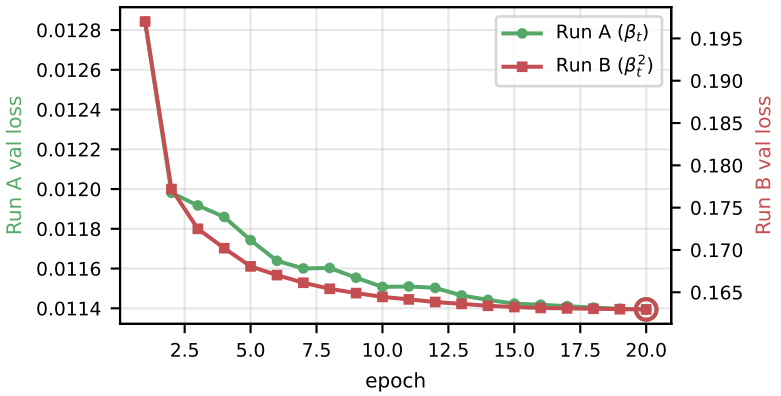
Noise-Schedule Ablation in DDPM: Photo -> MetFaces Portrait Translation

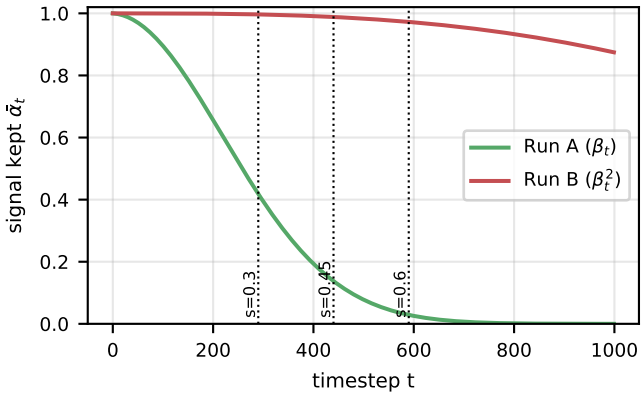
Key points

- Setup: google/ddpm-celebahq-256 fine-tuned on MetFaces (1069 train / 134 val / 133 test), 20 epochs. Run A and Run B share every hyper-parameter (lr 2e-05, cosine schedule, EMA 0.999, effective batch 8, seed 42); only the noise schedule differs.
- Training converged: Run A validation loss 0.01284 (epoch 1) -> 0.01139 (epoch 20), 11.3% lower; Run B 0.1970 -> 0.1630 (epoch 20), 17.3% lower. Best-validation checkpoints are used for testing.
- A and B losses are on different scales and must not be compared directly: under beta_t^2 the model must predict a tiny noise component from an almost clean image, which is intrinsically harder (higher MSE).
- At strength 0.45, Run A (212.1) does not beat Run 0 (109.0) on KID x10^3.
- Run B stays photographic: KID x10^3 181.8 vs 248.9 for the untouched photos, LPIPS to input 0.223 (Run A 0.625). Cause (chart b): with beta_t^2, alpha_bar_T = 0.87, so even full noising keeps 87% of the photo; at s=0.45 (t=440) the input keeps 99% of its signal under beta_t^2 vs 14% under beta_t, leaving the reverse process nothing to repaint.
- Strength is a style/identity trade-off (chart d): higher s lowers KID but raises LPIPS; the same s and seed are used for all three runs, so the comparison is fair at every point.
- Evaluation: 30 unseen test photos; KID (unbiased for small samples, error bars = std over subsets) against the 133 held-out MetFaces test images; LPIPS (AlexNet) to each input.

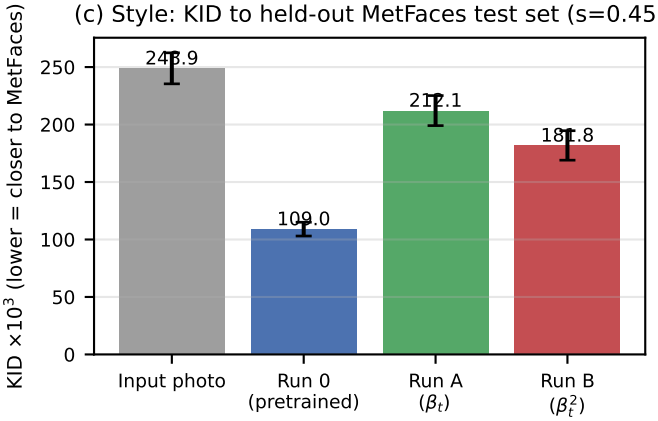
(a) Validation loss (fixed noise draw; separate axes)



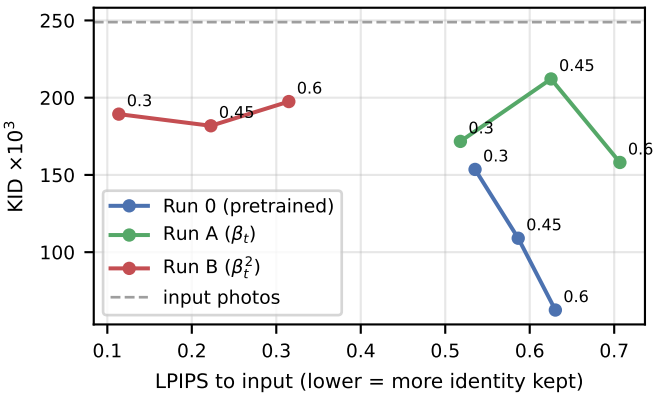
(b) Signal left after noising: β_t vs β_t^2



(c) Style: KID to held-out MetFaces test set (s=0.45)



(d) Style vs identity across strengths s



(e) Test photos and translations (strength s=0.45, same seed for all runs)

